

Refraction Mesh Architecture: Multimodal Decentralized Inference with Verifiable Claims and Fail-Closed Data Protection

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Abstract—Foundation models concentrate two coupled risks that existing assistant architectures handle poorly: *epistemic risk*, in which fluent generation outruns grounded verification, and *custody risk*, in which multimodal case material is retained inside a monopolistic compute boundary. Automated fact-checking and retrieval-augmented generation improve coverage, yet typically collapse judgment into a single score produced inside a single trust domain. We propose the *Refraction Mesh Architecture* (RMA), a systems architecture in which composite multimodal inputs are refracted into typed atomic claims and calculation tasks, probed by orthogonal witness planes, and admitted to export only through a machine-checkable publish gate. We formalize spectral conflict intensity κ_i , orthogonality Ω , gate predicates Γ , and a six-layer data-protection stack with fail-closed semantics. RMA couples ProofPath audit packets, privacy sleeves, dual decentralized/centralized routing with explicit provenance, and Proof-of-Inference cosigning. We situate the design against Byzantine consensus, mix networks, federated privacy, FEVER-style verification, and multi-agent debate, and argue that RMA’s novelty is compositional rather than a single new learning algorithm.

Index Terms—Decentralized inference, multimodal verification, publish gate, privacy sleeves, Proof-of-Inference, fact checking, systems security

I. INTRODUCTION

The rapid deployment of large language and multimodal models has changed drafting practice across journalism, law, compliance, and open-source intelligence [1]–[3]. The scientific and operational problem is no longer merely “can the model answer?” but “under what architectural guarantees may an answer become a publishable artifact, and what happens to the underlying media while that answer is produced?”

Empirical work shows that false claims propagate faster than true ones in online networks [4]. Analyses of large generative systems warn that fluency can detach from communicative grounding [5]. Instruction tuning and preference optimization improve helpfulness [3], yet do not by themselves create multi-party epistemic checks, durable negative evidence, or custody boundaries for restricted images and sealed documents.

Automated fact-checking has matured into extract–retrieve–verify pipelines with shared benchmarks [6]–[8]. Retrieval-augmented generation (RAG) reduces some hallucinations by coupling parametric memory to external corpora [9]. Chain-of-thought and multi-agent setups improve intermediate structure [10], [11]. Vision–language models enable captioning and

visual question answering [12], [13]. Journalism verification handbooks, however, insist on practices that most model UIs still omit: independent corroboration, documentation of what was *not* found, and human authority over publication [14], [15].

We therefore treat verification and data protection as *architectural invariants*. The central metaphor of this paper—refraction—is not decorative. A composite statement (prose plus pixels plus tables) behaves like white light: many frequencies superimposed. RMA refracts that mixture into typed spectral lines, measures disagreement across orthogonal instruments (witness planes), and applies a notch filter (publish gate) before the reconstituted signal is allowed to leave the desk.

A. Research Questions

- **RQ1 (Epistemics):** How can multimodal claims be certified without collapsing to a single model score inside a single trust domain?
- **RQ2 (Decentralization):** How can inference remain decentralized-first while still offering labeled centralized capacity when necessary?
- **RQ3 (Custody):** How can sensitive multimodal data be protected in motion and at rest under fail-closed policy rather than best-effort redaction?

B. Contributions

- 1) We invent and specify the Refraction Mesh Architecture for typed multimodal claim verification.
- 2) We formalize spectral conflict κ_i , plane orthogonality Ω , and publish-gate predicates Γ .
- 3) We define ProofPath packets with first-class negative evidence and Merkle–Ed25519 sealing [16], [17].
- 4) We present a dual inference fabric and a rigorously layered data-protection stack with fail-closed semantics [18]–[20].
- 5) We provide an evaluation agenda and a deployment mapping to a live network implementation.

II. RELATED WORK

A. Fault Tolerance, Ledgers, and Content Addressing

The Byzantine generals problem establishes that agreement among unreliable parties requires quora [21]. Practical Byzan-

tine Fault Tolerance (PBFT) made such quora operational for state-machine replication [22]. RMA borrows the *quorum* idea but applies it to epistemic artifacts—claims—rather than only to replica logs. Merkle trees enable compact commitments to structured objects [16]. Bitcoin and subsequent ledgers popularized append-only settlement of digests [23], [24]. Content-addressed networks motivate immutable extracts for later replay [25]. Public-key foundations and efficient signatures make packet verification cheap for desks [17], [26].

B. Privacy, Mix Networks, and Collaborative Learning

Mix networks unlink senders from messages by batching and reordering [18]. Differential privacy formalizes disclosure risk under neighboring datasets [27]. Secure aggregation and federated learning reduce central visibility into raw client updates [19], [20]. Interactive proofs motivate asymmetric verification cost [28]. RMA adapts unlinkability to inference *tasks*: the compute node should learn the job, not the requester’s network identity, and restricted media should never silently become vendor training fuel.

C. Fact Checking, RAG, Agents, and Multimodality

FEVER formalized large-scale claim verification [6]; surveys chart the field’s methods and limits [7]. Contrastive evidence improves robustness to superficial cues [8]. RAG couples generation to retrieved non-parametrics [9]. Multi-agent systems explore debate and tool use [10], [11]. Multimodal LLMs fuse vision and language [1], [12], [13]. RMA is complementary: it does not claim a new backbone model; it claims a new *control plane* for when generative output may leave a high-stakes desk.

III. THREAT MODEL AND DESIGN GOALS

We assume a minority of compute nodes may be Byzantine (lie, delay, collude) [21]. Centralized model APIs may be reachable but are untrusted for sole judgment and for restricted multimodal case files [2], [5]. Adversaries may attempt prompt injection, source poisoning, Sybil “witnesses,” UI exfiltration, adversarial images, OCR spoofing, and deliberate caption–pixel inconsistency [13].

a) Design goals.:

- 1) **G1 Multiplicity**: no single plane is sufficient for publication-bound claims.
- 2) **G2 Durability**: intermediate checks survive as replayable packets, not ephemeral UI traces.
- 3) **G3 Custody**: restricted media defaults to on-node routes; catalog use is explicit and logged.
- 4) **G4 Fail-closed**: absence of private capacity refuses the job rather than degrading into exfiltration.
- 5) **G5 Human authority**: gates assist editors; they do not replace them [14], [15].

Non-goals include perfect truth discovery, anonymity against global passive network adversaries beyond sleeve relays, and training frontier foundation models.

IV. REFRACTION MESH ARCHITECTURE

A. Optical Metaphor as Systems Invariant

Let a multimodal bundle be $B = (Q, R, \mathcal{M})$, where Q is the query, R a generative reply, and $\mathcal{M} = \{m_j\}$ attachments (images, PDFs, tables, optional transcripts). *Refraction* is a map

$$\text{Refract}(B) \mapsto C = \{c_1, \dots, c_n\}, \quad (1)$$

where each c_i is a typed atom. Disagreement across instruments is measured; contested frequencies are attenuated by the publish gate before export.

B. Typed Atoms

Each atom carries a type

$$\tau(c_i) \in \{\text{text, vision, doc, table, calc, cross}\}. \quad (2)$$

calc atoms are propositions about arithmetic or unit relations that should be checked by deterministic procedures, not only by another language model. cross atoms assert consistency across modalities (e.g., “the figure shows \$12M” must match OCR/table and prose).

C. Witness Planes

A witness plane is a function

$$W_k : (c_i, \mathcal{S}) \mapsto v_{i,k} \in \{\text{support, contest, unknown}\}, \quad (3)$$

given source set $\mathcal{S}$. RMA requires the following families for publication-bound work:

- 1) Record plane (primary documents);
- 2) Temporal plane (chronology / EXIF constraints);
- 3) Compute plane (diverse weights or nodes);
- 4) Contradiction plane (active disconfirmation) [8];
- 5) Modality plane (cross-modal consistency) [12].

D. Spectral Conflict Intensity

Let

$$s_i = |\{k : v_{i,k} = \text{support}\}|, \quad t_i = |\{k : v_{i,k} = \text{contest}\}|. \quad (4)$$

Define spectral conflict intensity

$$\kappa_i = \frac{t_i}{s_i + t_i + \varepsilon}, \quad \varepsilon > 0. \quad (5)$$

Under threshold $\tau_\kappa \in (0, 1)$, claim c_i is contested if $\kappa_i \geq \tau_\kappa$, or if any required plane returns contest when desk policy is strict.

E. Packet Object

A ProofPath packet is the tuple

$$P = (Q, R, \mathcal{M}_{\text{handles}}, C, \{(W_k, v_{i,k})\}, \mathcal{N}, G, \delta, \sigma), \quad (6)$$

where $\mathcal{M}_{\text{handles}}$ are content hashes (not necessarily raw bytes), $\mathcal{N}$ is the negative-evidence log, G the gate decision, $\delta = \text{MerkleRoot}(P \setminus \{\delta, \sigma\})$ [16], and σ a set of Ed25519 cosignatures [17].

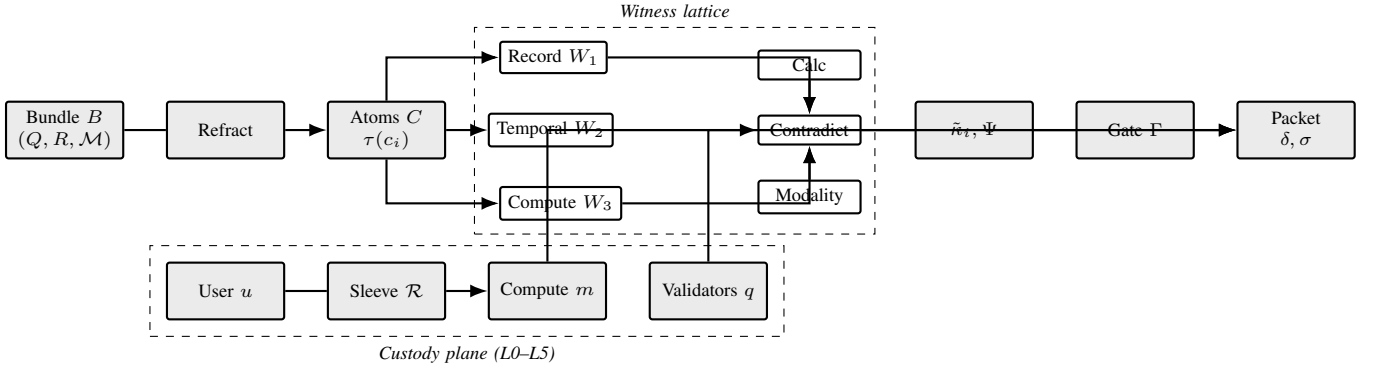


Fig. 1. Refraction Mesh: witness lattice (top) coupled to sleeve/custody plane and cosigned packet sealing (bottom).

F. Orthogonality

Let $E_k \in \{0,1\}$ be plane error indicators on a labeled contested set. Empirical orthogonality is

$$\Omega = 1 - \frac{2}{m(m-1)} \sum_{1 \leq k < \ell \leq m} \text{Corr}(E_k, E_\ell). \quad (7)$$

Low Ω diagnoses collapsed diversity (e.g., all compute planes calling one provider) [5].

G. Weighted Spectral Measure and Soft Gate Score

Not all planes are equal. Let $w_k > 0$ be plane weights with $\sum_k w_k = 1$. Define weighted support/contest mass

$$S_i = \sum_{k:v_{i,k}=\text{support}} w_k, \quad T_i = \sum_{k:v_{i,k}=\text{contest}} w_k, \quad U_i = \sum_{k:v_{i,k}=\text{unknown}} w_k. \quad (8)$$

The weighted conflict is

$$\kappa_i^{(w)} = \frac{T_i}{S_i + T_i + \varepsilon}, \quad \tilde{\kappa}_i = \kappa_i^{(w)} (1 - \alpha U_i), \quad (9)$$

where $\alpha \in [0,1]$ down-weights conflict when ignorance dominates (unknown-heavy claims should tend to review, not false ready).

A soft publish score aggregates atoms:

$$\Psi(P) = \exp\left(-\beta \sum_{i=1}^n \tilde{\kappa}_i\right) \cdot \mathbf{1}\left[\min_i |\mathcal{W}_i| \geq q_{\min}\right], \quad (10)$$

with temperature $\beta > 0$. Hard gate Γ can be recovered by thresholding Ψ and applying the strict contest rule.

H. Information Bound for Sleeve Unlinkability

Let U be requester identity, M the multimodal payload, L_m compute logs, and R the relay transcript. Under an honest-but-curious compute node and a correctly functioning sleeve,

$$I(U; L_m) \leq I(U; R) + I(U; M | R). \quad (11)$$

Private mode stores task hashes at the hub and strips transport identifiers at $\mathcal{R}$, driving the first term toward zero for local adversaries. The second term is controlled by minimization (handles): desks SHOULD ensure M itself does not embed identifying metadata (EXIF GPS, named filenames).

I. Consistency Functional Across Modalities

For a cross-atom c linking modalities a, b , define a consistency functional

$$\phi(c) = 1 - \frac{\|e_a(c) - e_b(c)\|_2}{\|e_a(c)\|_2 + \|e_b(c)\|_2 + \varepsilon}, \quad (12)$$

where e_a, e_b are embeddings or normalized numeric feature maps (OCR totals, caption vectors, table aggregates). Plane W_{modality} returns contest if $\phi(c) < \tau_\phi$.

V. MULTIMODAL CALCULATION PLANE

Multimodal desks do not only chat; they calculate. Let Calc be a deterministic interpreter over extracted numeric relations:

$$\text{Calc}(c_i) \mapsto \{\text{match}, \text{mismatch}, \text{n/a}\}. \quad (13)$$

We fold calc outcomes into spectral conflict by treating mismatch as a contesting vote. This is a deliberate architectural choice: language models are strong at fluency and weak at reliable arithmetic consistency with tables [1], [5]. A calc veto is cheaper and more auditable than hoping a second model “notices.”

a) Pipeline.: (1) content-address attachments; (2) run modality specialists (OCR, table parse, VQA) preferably on-node for restricted media; (3) refract into typed atoms; (4) schedule witness+calc planes; (5) log negative evidence; (6) apply Γ ; (7) seal (G, δ, σ) . Streaming tokens remain a latency surface; sealed packets are the product of record.

VI. DECENTRALIZED DUAL FABRIC

A. Principle

Decentralized-first, centralized-explicit. Self-hostable weights and independent nodes are the sovereignty path. Centralized catalog routes may expand coverage but must never masquerade as local compute: both the UI and the packet label the route.

B. Roles and Settlement

Let roles be hub $\mathcal{H}$ (discovery, validator keys), relay $\mathcal{R}$ (sleeve hop), compute $\mathcal{C}$, witness feeds $\mathcal{F}$, and desk $\mathcal{D}$.

After inference, an attestation enters a Proof-of-Inference log. Validators produce signatures; network-finality requires

$$|\sigma| \geq q, \quad (14)$$

for configurable quorum q , in the spirit of Byzantine quora [22] but scoped to inference receipts rather than full SMR [23].

C. Why Multimodality Raises the Stakes

Images and sealed PDFs are high-value exfiltration targets. Defaulting restricted multimodal inference to on-node routes reduces vendor retention surface area. When a desk must use a centralized vision route, RMA requires: explicit labeling, packet provenance, optional sleeve, and never silent fallback.

VII. DATA PROTECTION STACK

Product UIs often reduce “privacy” to a banner or a retention FAQ. RMA instead treats custody as a layered protocol with explicit failure modes. Let sensitivity class $\lambda(B) \in \{\text{public}, \text{internal}, \text{restricted}\}$.

A. Layer L0 — Classification

Before routing,

$$\rho(B) = \begin{cases} \text{on-node,} & \lambda(B) = \text{restricted,} \\ \text{prefer-node,} & \lambda(B) = \text{internal,} \\ \text{any-labeled,} & \lambda(B) = \text{public.} \end{cases} \quad (15)$$

Classification errors are themselves security bugs: over-labeling reduces utility; under-labeling violates G3.

B. Layer L1 — Privacy Sleeves (Data in Motion)

Requester u submits ciphertext task $\tilde{T}$ to hub $\mathcal{H}$. Relay $\mathcal{R}$ strips transport identity and forwards to compute $m \in \mathcal{C}$ [18]. Idealized unlinkability targets near-zero mutual information

$$I(U; L_m) \approx 0 \quad (16)$$

between requester identity U and compute-side logs L_m under private mode, acknowledging that global passive adversaries remain out of scope.

C. Layer L2 — Minimization and Handles

Sealed packets store handles $h(m_j) = \text{Hash}(m_j)$ and excerpts, not entire archives [25]. Raw blobs remain in desk-controlled storage with access control. This separates *auditability* from *retention of pixels*.

D. Layer L3 — Encryption and Retention (Data at Rest)

Desk artifacts (synced chats, exports) SHOULD be encrypted under desk keys. Let retention clock $\mathcal{T}$ map artifact class to time-to-live. Negative-evidence rows may out-live raw media when policy requires audit without keeping biometrics—a deliberate asymmetry.

Algorithm 1 Multimodal ProofPath (simplified)

Require: $Q, \mathcal{M}, \Pi, \mathcal{S}$

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1: classify  $\lambda$ ; store handles  $h(m_j)$ 
2: if  $\lambda = \text{restricted}$  and  $\neg A_{\text{private}}$  then
3:   refuse ▷ L5 fail-closed
4: end if
5:  $R \leftarrow \text{Infer}(Q, \mathcal{M})$  with labeled route  $r$ 
6:  $C \leftarrow \text{AtomizeTyped}(R, \mathcal{M})$ 
7: for all  $c_i \in C$  do
8:   collect  $v_{i,k} \leftarrow W_k(c_i, \mathcal{S})$  incl. calc/modality
9:   append contradiction + negative-evidence rows to  $\mathcal{N}$ 
10: end for
11:  $G \leftarrow \Gamma(P, \Pi)$ ;  $\delta \leftarrow \text{MerkleRoot}(\cdot)$ ; solicit  $\sigma$ 
12: return sealed  $P$ 

```

E. Layer L4 — Route Provenance and UI Honesty

Every reply and packet records route label

$$r \in \{\text{decentralized/on-node, centralized/catalog}\}. \quad (17)$$

Silent cross-cloud offload is a protocol violation even if the answer is “correct.” Honesty here is part of security, not marketing.

F. Layer L5 — Fail-Closed Private Desks

Define availability predicate A_{private} . If $\lambda(B) = \text{restricted}$ and $\neg A_{\text{private}}$, then

$$\text{Accept}(B) = \text{false}. \quad (18)$$

The system must refuse rather than “helpfully” send pixels to a public catalog. Fail-open assistants optimize engagement; fail-closed desks optimize non-exfiltration.

G. What This Stack Is Not

Sleeves are not a substitute for endpoint security. Screenshots bypass packet controls. Differential privacy is not claimed for generative outputs [27]. The contribution is a coherent custody control plane aligned with high-stakes desk practice.

VIII. PROOFPATH PROTOCOL

ProofPath realizes RMA as an operational spiral:

- 1) Atomize typed claims from $(R, \mathcal{M})$.
- 2) Witness across required planes.
- 3) Contradict with forced disconfirmation.
- 4) Log negative evidence (empty searches are first-class) [14].
- 5) Persist extracts for third-party replay.
- 6) Apply publish gate Γ .

IX. PUBLISH GATE FORMALISM

Desk policy $\Pi = (\tau_\kappa, q_{\min}, \mathcal{W}_{\text{req}}, \text{strict})$ induces

$$\Gamma(P, \Pi) \in \{\text{ready}, \text{review}, \text{blocked}\}. \quad (19)$$

With $\mathcal{W}_i$ the set of planes that voted on c_i ,

$$\text{blocked} \Leftarrow \exists c_i : \kappa_i \geq \tau_\kappa \wedge \text{strict}, \quad (20)$$

$$\text{ready} \Leftarrow \forall c_i : \kappa_i < \tau_\kappa \wedge |\mathcal{W}_i| \geq q_{\min}, \quad (21)$$

$$\text{review} \Leftarrow \text{otherwise}. \quad (22)$$

Export formats refuse blocked packets by default. This encodes editorial caution as a predicate rather than as social hope [15].

X. SECURITY ANALYSIS

a) Sybil witnesses.: If anonymous web hits can cast gate-decisive votes, an adversary floods support. Therefore gate-decisive planes must be keyed to stake, institutional keys, or desk allowlists; raw web evidence remains advisory.

b) Model collusion.: If all compute planes call one provider, orthogonality Ω collapses [2]. Diversity labels and on-node options are mandatory for publish-critical work.

c) Poisoned multimodal evidence.: Adversarial images and OCR spoofing are contested via modality and contradiction planes; calc mismatches fail closed under strict Π .

d) Privacy leakage.: Sleeves reduce linkability [18]; extreme desks keep private routing fully on-prem [20].

XI. NOVELTY AND POSITIONING

Scientific novelty in systems work is often compositional: a new arrangement of known primitives that enables new guarantees. RMA's claim is of that kind—not a new backbone model, but a new control plane. We state eight numbered positioning claims:

- N1 Typed multimodal atoms.** Unlike single-score chat and limited FEVER extractors [6], [7], RMA refracts bundles into $\tau(c_i) \in \{\text{text}, \text{vision}, \text{doc}, \text{table}, \text{calc}, \text{cross}\}$.
- N2 Deterministic calc veto.** Unlike RAG/LLM stacks that hope a second model notices arithmetic errors [1], [9], Calc can increment t_i via mismatch.
- N3 Orthogonal witness quora.** Unlike corpus-only checking [6], publication requires multi-plane votes with measurable orthogonality Ω (Eq. (7)).
- N4 Negative evidence as schema.** Unlike ephemeral UI traces, empty searches are first-class rows in $\mathcal{N}$ [14].
- N5 Export-blocking gate Γ .** Unlike soft scores, $\Gamma \in \{\text{ready}, \text{review}, \text{blocked}\}$ can refuse export under $\kappa_i \geq \tau_\kappa$ [15].
- N6 Durable cosigned packets.** Unlike multi-agent debate that discards intermediate turns [10], [11], RMA seals (G, δ, σ) for replay.
- N7 Structured epistemic notarization.** Unlike ledger blob timestamps alone [23], [24], cosigning covers claims, votes, absences, and gate—not only existence at time t .
- N8 Fail-closed custody protocol.** Unlike “private chat” product modes, layers L0–L5 encode classification, sleeves [18], handles, encryption, route honesty, and $\text{Accept}(B) = \text{false}$ when private capacity is missing.

XII. WORKED EXAMPLE

Consider prose asserting that “Mayor Reyes diverted \$12M of transit funds to a private foundation in March 2025,” accompanied by a budget PDF page image. Refraction yields c_1 (amount), c_2 (date), c_3 (destination), c_4 (attribution), and a cross/calc atom linking OCR totals to prose. If the temporal plane contests March while calc matches \$12M, then κ_2 rises. Under strict Π , Γ returns blocked until the date is resolved. An empty FOIA index query enters $\mathcal{N}$ —absence as evidence [14].

XIII. IMPLEMENTATION AND EVALUATION AGENDA

The Noeti deployment maps RMA onto a discovery hub, Chat and spatial Canvas surfaces, ProofPath Desk, on-node open weights, a labeled network catalog, and live wire feeds. Proposed metrics include gate precision/recall on journalist-labeled contested claims [6]; replay fidelity of δ ; orthogonality Ω ; sleeve efficacy $I(U; L_m)$; calc-plane catch rate; and restricted-mode refusal rate when only centralized vision routes exist.

XIV. LIMITATIONS AND ETHICS

RMA structures disagreement; it does not mint truth [15]. Over-confident gates can suppress legitimate contested reporting; under-confident gates fail the beachhead. Human editors retain authority [14]. If pixels leave the desk, the packet must say so—otherwise privacy UI becomes security theatre.

XV. CONCLUSION

We presented the Refraction Mesh Architecture as a research systems design for multimodal, decentralized, custody-aware claim verification. By refracting composites into typed atoms, measuring spectral conflict across orthogonal planes, protecting data through a fail-closed stack, and sealing publish decisions under Proof-of-Inference cosignatures, RMA offers high-stakes desks a path to use generative models without surrendering either epistemic standards or case files to a single cloud.

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TABLE I

TRACTABLE MEASURES OF VERIFICATION AND CUSTODY (PHYSICAL/OPERATIONAL UNITS). BIT = INDICATOR IN $\{0, 1\}$. UNBOUNDED MI = NO SLEEVE ARCHITECTURE. RMA ENTRIES ARE INVARIANTS OR EVALUATION TARGETS.

Measure (unit)	LLM chat	FEVER/RAG	RMA
Witness planes for publish m (count)	0	1	≥ 2
Typed atom families $ \mathcal{T} $ (count)	1	1	6
Calc $\rightarrow$ contest coupling (bit)	0	0	1
Neg.-evidence rows $ \mathcal{N} $ if searched (count)	0	0	≥ 1
Export block if $\kappa_i \geq \tau_\kappa$ (bit)	0	0	1
Merkle digest $ \delta $ (bits)	0	0	256
Cosign quorum q (signatures)	0	0	≥ 1
Route label on reply (bit)	0	0	1
Refuse if $\neg A_{\text{private}}$ (bit)	0	0	1
Sleeve $I(U; L_m)$ (nats, private)	unbdd.	unbdd.	$\rightarrow 0$
Orthogonality Ω (dimensionless)	n/a	~ 0	$\uparrow 1$
Replay fidelity of δ (%)	n/a	n/a	≥ 99
Warm ProofPath T_{90} (s)	n/a	n/a	≤ 90
False-ready @ strict II (%)	n/a	n/a	< 5

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